

# Watt-Wise

## SMART ENERGY MONITORING AND WASTE DETECTION SYSTEM

| NAME                | ROLL NUMBER |
|---------------------|-------------|
| Y. SRI ANJANA       | 2420030204  |
| DEVANABOINA LIKITHA | 2420030295  |
| CH. HEMAVATHI       | 2420030745  |

*UNDER THE GUIDANCE OF*

Dr. N. SHIRISHA  
(ASSISTANT PROFESSOR)

# Title Of Contents

---



1. Introduction
2. Problem Statement
3. Abstract
4. Objectives
5. Existing Methodology Vs Proposed Methodology
6. Literature Review & Gap
7. Technology Stack

# Abstract

---



**WattWise** is a Big Data-based smart energy monitoring system designed to identify abnormal electricity consumption and potential energy wastage in buildings such as colleges, offices, and institutions. The system uses **simulated smart-meter data** to continuously generate energy-consumption records without requiring physical IoT hardware. The data is ingested through **Apache Kafka**, stored in **Hadoop HDFS**, and processed using **Apache Spark** for data cleaning, transformation, and analysis. An **anomaly detection module** identifies unusual consumption patterns that may indicate inefficient or unnecessary energy usage. The processed data and detected anomalies are visualized through a **Power BI dashboard**, providing consumption trends, high-usage areas, and actionable insights. Thus, WattWise provides an integrated Big Data pipeline for **continuous energy monitoring, anomaly detection, and data-driven energy management**.

# Introduction

---



- **Growing Energy Consumption:** Buildings such as colleges, offices, and institutions consume large amounts of electricity for daily operations.
- **Energy Wastage:** Unnecessary usage, inefficient equipment, and abnormal consumption patterns can result in significant energy wastage.
- **Monitoring Challenges:** Conventional monitoring methods may rely on manual observation or periodic readings, making it difficult to identify unusual consumption promptly.
- **Need for Continuous Monitoring:** Continuous collection and analysis of electricity-consumption data can help identify abnormal usage patterns more effectively.
- **Role of Big Data:** Large volumes of energy data require scalable technologies for **data ingestion, storage, processing, and analysis**.
- **Anomaly Identification:** Analyzing consumption patterns can help detect unusual energy usage that may indicate potential wastage.
- **Data-Driven Decisions:** Visualizing consumption trends and anomalies can help organizations understand energy usage and make informed decisions to improve energy efficiency.



# Problem Statement:

---



To develop a scalable system that continuously analyses electricity-consumption data, detects abnormal usage patterns, and provides actionable insights for reducing energy wastage.

# Literature Survey

| Study (Author, Year)                                                     | References                                                                                                                                                    | Focus                                                                             | Technique / Approach                                                              | Limitation                                                                        |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Hajjaji et al. (2021) Computer Science Review                            | <a href="https://www.sciencedirect.com/science/article/abs/pii/S1574013720304184">https://www.sciencedirect.com/science/article/abs/pii/S1574013720304184</a> | Systematic review of IoT + Big Data applications across smart-environment domains | PRISMA-based review of 45 peer-reviewed studies (2011-2019)                       | Broad, general-purpose survey; no building-specific wastage-detection pipeline    |
| EL-Hadad, Tan & Tan (2022) Int'l Journal of Technology                   | <a href="https://ijtech.eng.ui.ac.id/article/view/5931">https://ijtech.eng.ui.ac.id/article/view/5931</a>                                                     | Predicting abnormal electricity consumption from smart-meter readings             | Isolation Forest for labeling, then Random Forest / Decision Tree for forecasting | Trained and tested on static historical datasets, not a live data stream          |
| Ansari, Vakili & Bahrak (2019) Journal of Big Data                       | <a href="https://link.springer.com/article/10.1186/s40537-019-0270-8">https://link.springer.com/article/10.1186/s40537-019-0270-8</a>                         | Benchmarking Big Data frameworks for smart-grid meter data                        | Simulated smart-meter generator, Kafka queue, NoSQL stores (HBase, Cassandra)     | Focused on database performance only; no anomaly detection or visualization layer |
| Sonawane & Pandit (2024) Int'l Journal of Research Publication & Reviews | <a href="https://ijrpr.com/uploads/V5ISUE1/IJRPR22184.pdf">https://ijrpr.com/uploads/V5ISUE1/IJRPR22184.pdf</a>                                               | Visualizing energy consumption trends for decision-making                         | Power Query + DAX measures feeding a Power BI dashboard                           | Visualization-only; not connected to any upstream Big Data pipeline               |

# Literature Survey

| Aspect            | Existing Approaches                                                                        | Gap                                                                         | Future Work                                                                  |
|-------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Data Pipeline     | Ansari et al. (2019) simulate smart-meter data with Kafka, but only to benchmark databases | No single pipeline connects ingestion, anomaly detection, and visualization | Combines Kafka + HDFS + Spark end-to-end, from simulated data to dashboard   |
| Anomaly Detection | EL-Hadad et al. (2022) apply Isolation Forest and ML models to static datasets             | Not designed to run on continuous, streaming consumption data               | Anomaly detection runs within the Spark-based streaming pipeline             |
| Visualization     | Sonawane & Pandit (2024) build Power BI dashboards for energy data                         | Dashboard is disconnected from any Big Data backend or live detection       | Power BI dashboard is fed directly by Spark-processed, anomaly-tagged data   |
| Deployment Scope  | Hajjaji et al. (2021) survey IoT + Big Data across broad smart-environment domains         | General-purpose; not tailored to institutional building energy use          | Purpose-built for colleges, offices, and institutions to target real wastage |

# Objectives

---



1. **Monitor** electricity consumption continuously using generated smart-meter data.
2. **Detect** potential energy wastage through anomaly detection.
3. **Visualize** consumption trends and anomalies for better understanding.
4. **Provide actionable insights** to support energy-efficient decision-making.



# Methodology

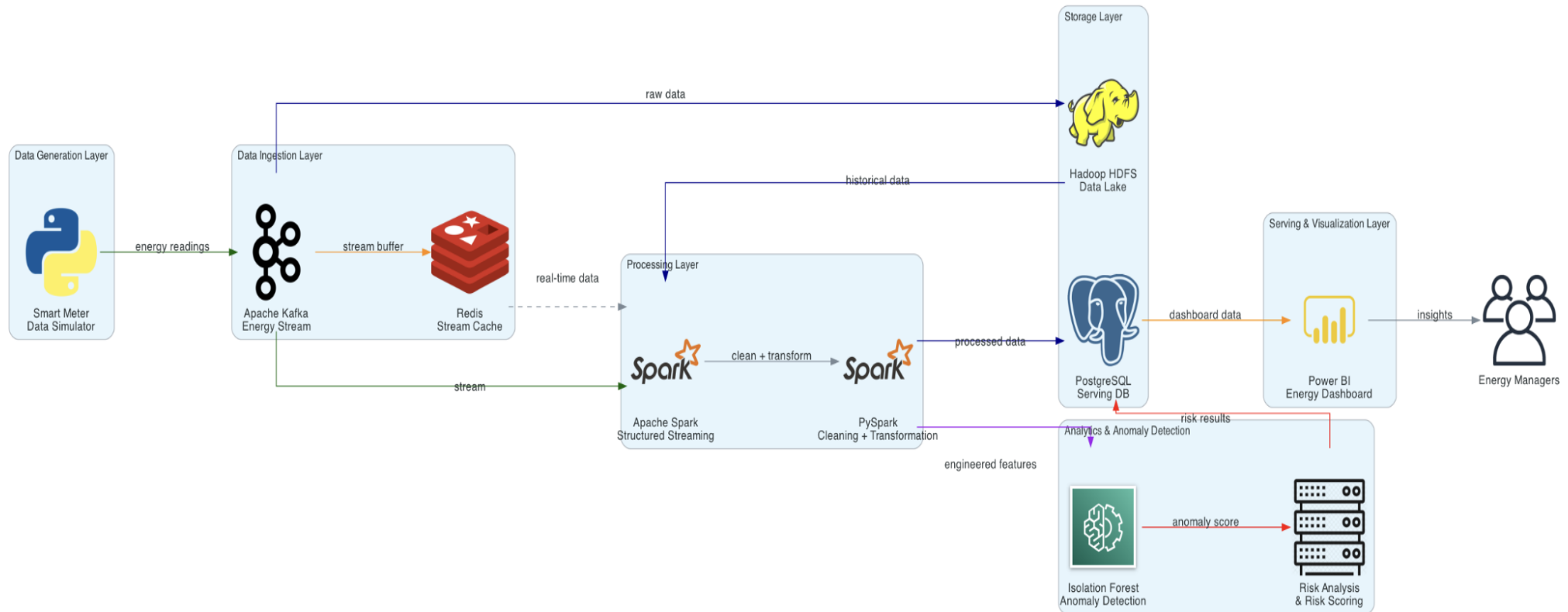
---

| EXISTING METHODOLOGY :                                                              | NEW METHODOLOGY :                                                                       |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Manual or periodic monitoring of electricity consumption.                           | Continuous generation and collection of smart-meter energy data.                        |
| Energy data is mainly analyzed from <b>static or historical records</b> .           | <b>Apache Kafka</b> is used for continuous data ingestion                               |
| Limited capability to handle <b>continuous and large-scale data</b> .               | <b>Apache Spark</b> performs data cleaning, transformation, aggregation, and analysis.  |
| Abnormal consumption is identified <b>after analysis rather than continuously</b> . | <b>Anomaly detection</b> identifies unusual energy-consumption patterns.                |
| Visualization is often <b>separate from the data-processing system</b> .            | <b>Power BI</b> provides interactive visualization of consumption trends and anomalies. |

# Technology Stack

| Layer               | Technology             | Purpose                             |
|---------------------|------------------------|-------------------------------------|
| Data Generation     | Python                 | Generate simulated smart-meter data |
| Data Ingestion      | Apache Kafka           | Stream energy data                  |
| Storage             | Hadoop HDFS            | Store raw and processed data        |
| Data Processing     | Apache Spark / PySpark | Clean, transform and process data   |
| Analytics           | Spark SQL              | Query and aggregate energy data     |
| Feature Engineering | Python / PySpark       | Create ML features                  |
| Machine Learning    | Scikit - learn         | Build anomaly Detection model       |
| Anomaly Detection   | Isolation Forest       | Identify abnormal consumption       |
| Risk Analysis       | Python                 | Calculate risk score                |
| Visualization       | Power BI               | Displays trends                     |

# System Architecture



WattWise - Smart Energy Monitoring Architecture